Introducción a la Física Clásica

Clase VI

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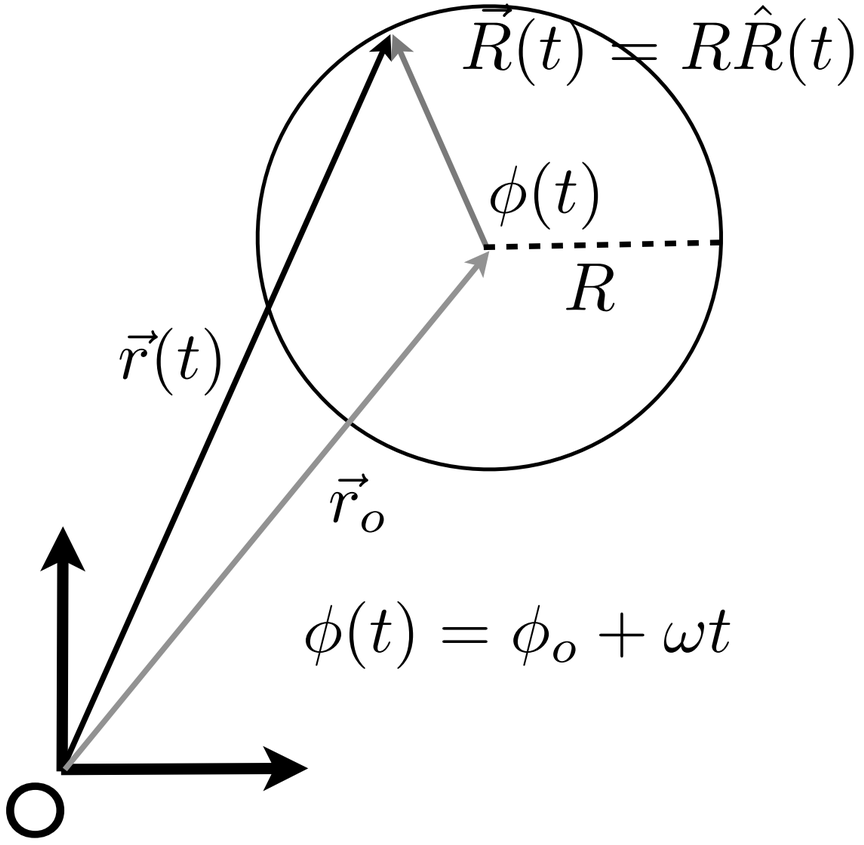
En la clase de hoy

Movimiento Circunferencial Acelerado

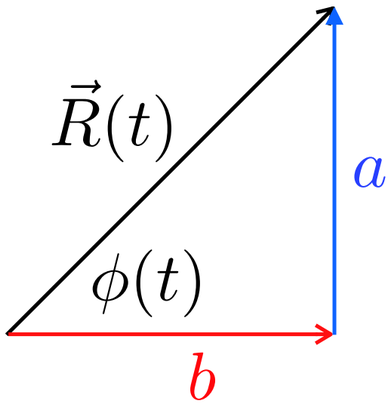
Circunferencial Uniforme

$\vec{r}(t)$: vector posición

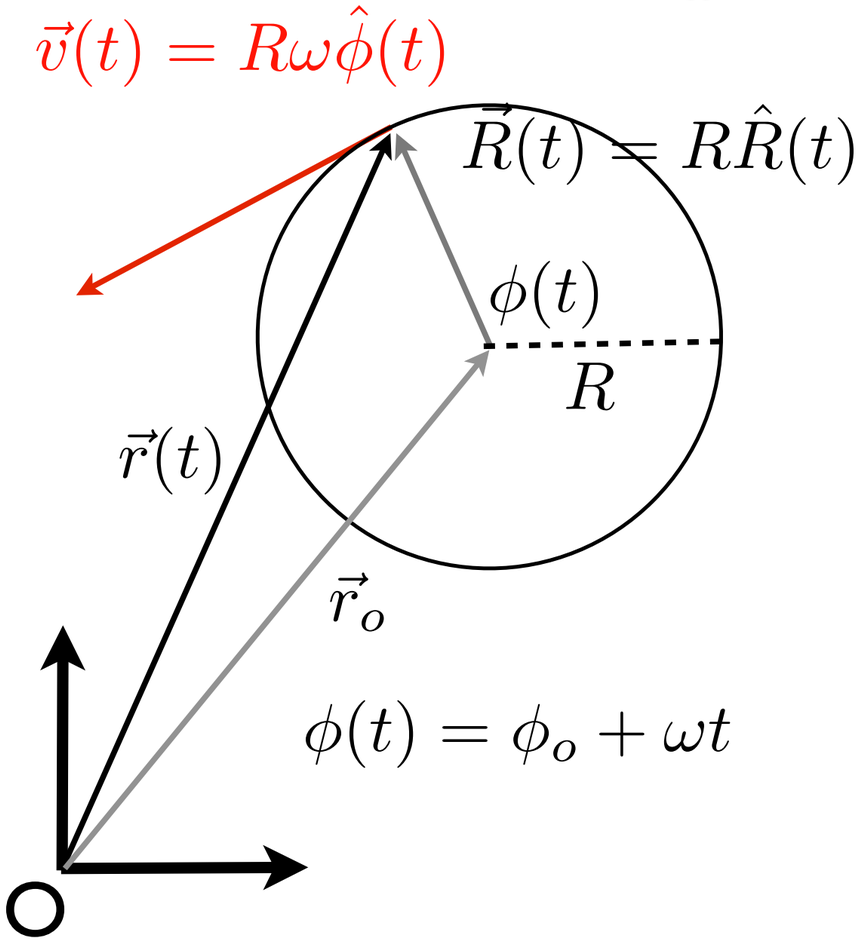


Para adonde apunta

$$\phi(t) = \phi_o + \omega t$$



Circunferencial Uniforme



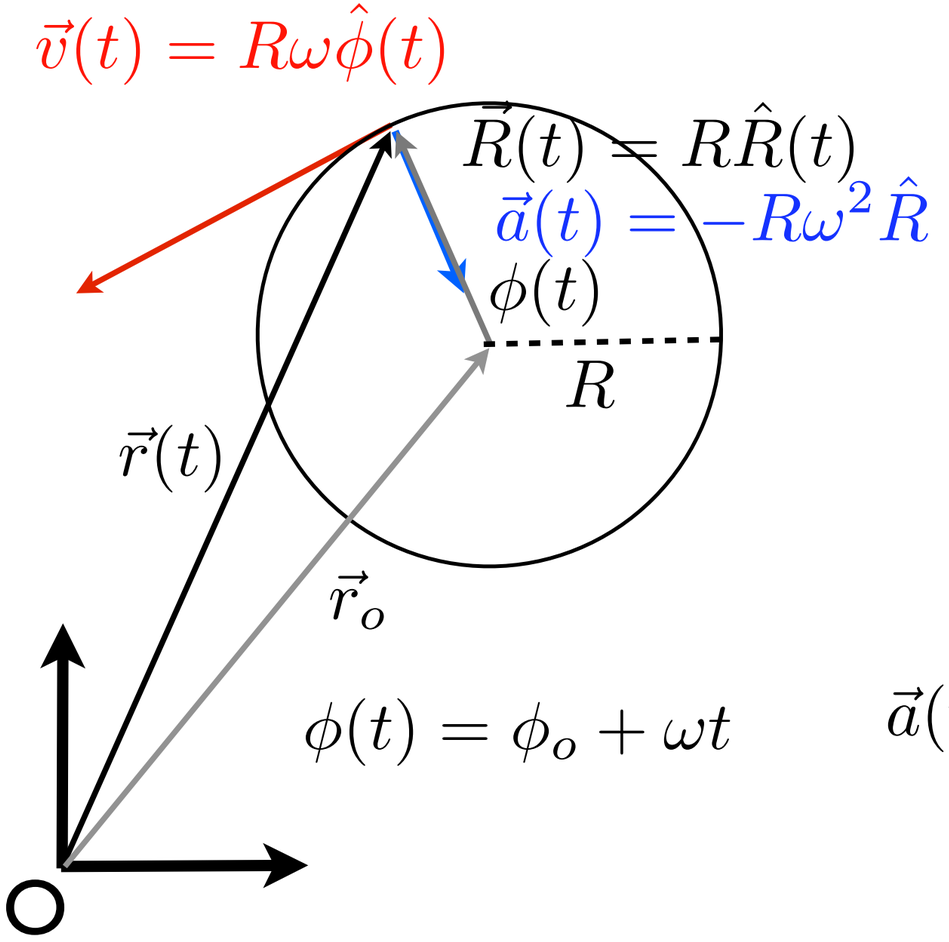
$\vec{r}(t)$: vector posición

$\vec{v}(t)$: vector velocidad

$$\phi(t) = \phi_o + \omega t$$

$$\vec{v}(t) = \lim_{\Delta t \to 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t}$$

Circunferencial



$\vec{r}(t)$: vector posición

$\vec{v}(t)$: vector velocidad

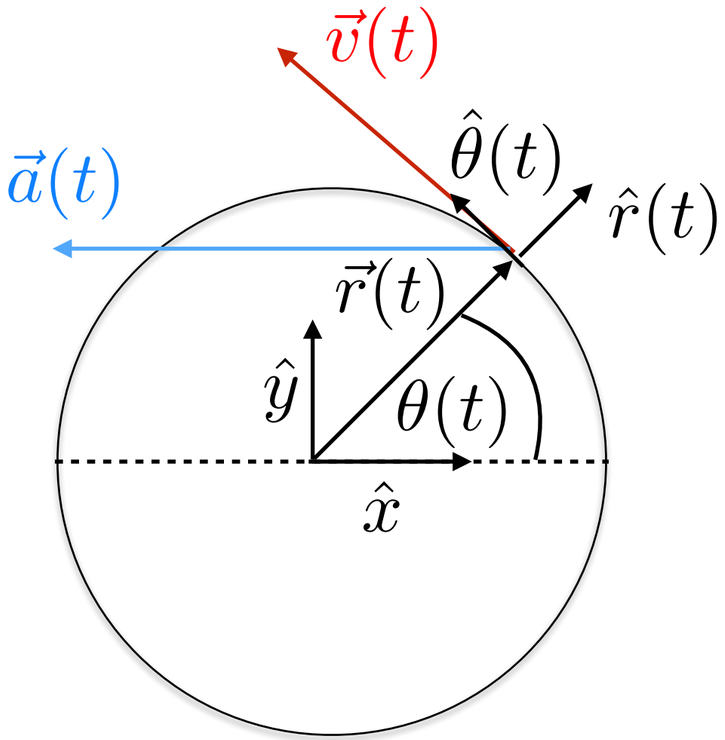
$\vec{a}(t)$: vector aceleración

$$\phi(t) = \phi_o + \omega t$$

$$\vec{a}(t) = \lim_{\Delta t \to 0} \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t}$$

Circunferencial Acelerado

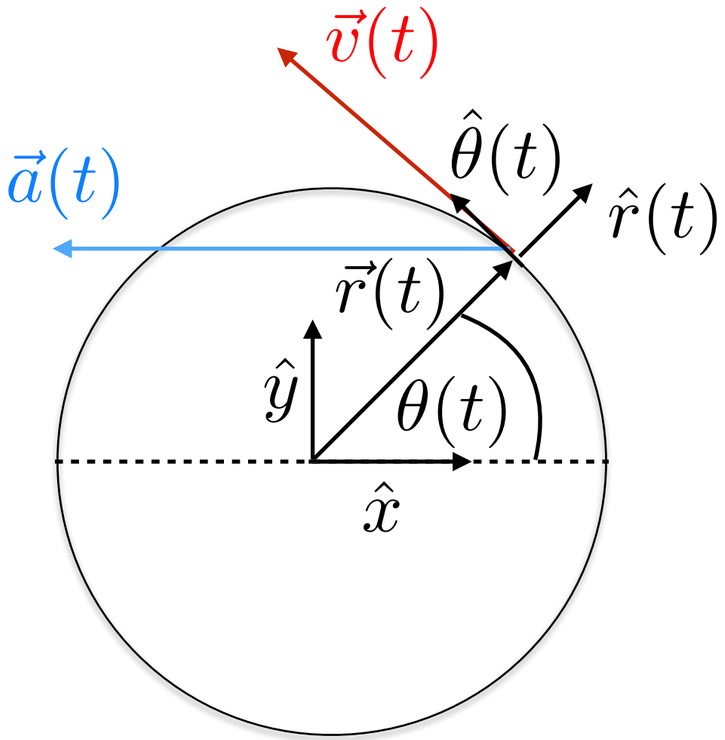
$$\theta_o + \omega_o t \rightarrow \theta(t)$$



$$\vec{r}(t) = R(\cos(\theta(t))\hat{x} + \sin(\theta(t))\hat{y})$$
$$= R\hat{r}(t)$$

Circunferencial Acelerado

$$\theta_o + \omega_o t \to \theta(t)$$



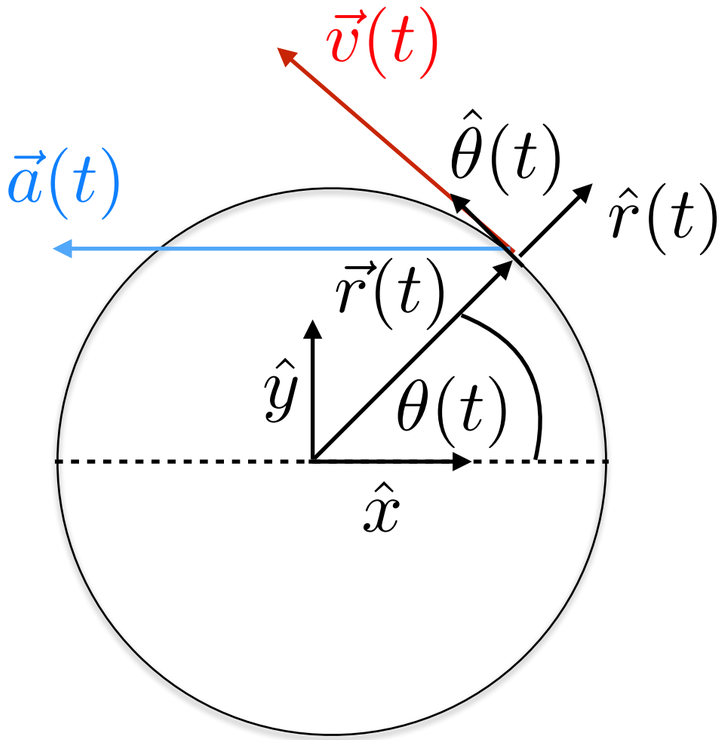
$$\vec{r}(t) = R(\cos(\theta(t))\hat{x} + \sin(\theta(t))\hat{y})$$
$$= R\hat{r}(t)$$

$$\vec{v}(t) = R\omega(t)(-\sin(\theta(t))\hat{x} + \cos(\theta(t))\hat{y})$$
$$= R\omega(t)\hat{\theta}(t)$$

Con $\omega(t) = \lim_{\Delta t \to 0} \frac{\theta(t+\Delta t) - \theta(t)}{\Delta t} \equiv \frac{d\theta(t)}{dt}$

Circunferencial Acelerado

$$\theta_o + \omega_o t \to \theta(t)$$



$$\vec{v}(t) = R\omega(t)(-\sin(\theta(t))\hat{x} + \cos(\theta(t))\hat{y})$$

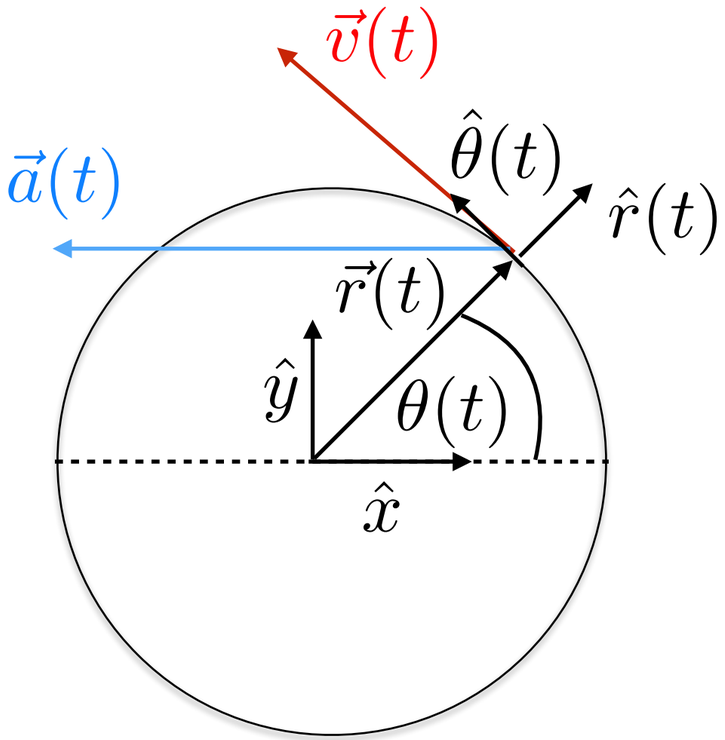
$$= R\omega(t)\hat{\theta}(t)$$

$$\vec{a}(t) = (-R\omega(t)^2\cos(\theta(t)) - R\alpha(t)\sin(\theta))\hat{x}$$
$$(-R\omega(t)^2\sin(\theta(t)) + R\alpha(t)\cos(\theta))\hat{y}$$

Con $\alpha(t) = \lim_{\Delta t \to 0} \frac{\omega(t+\Delta t) - \omega(t)}{\Delta t} \equiv \frac{d\omega(t)}{dt}$

Circunferencial Acelerado

$$\theta_o + \omega_o t \rightarrow \theta(t)$$



$$\vec{a}(t) = (-R\omega(t)^2 \cos(\theta(t)) - R\alpha(t)\sin(\theta))\hat{x}$$
$$(-R\omega(t)^2 \sin(\theta(t)) + R\alpha(t)\cos(\theta))\hat{y}$$

$$\vec{a}(t) = -R\omega(t)^2\hat{r}(t) + R\alpha(t)\hat{\theta}(t)$$

Aceleración radial

Aceleración tangencial